KRISHNA MURTHY CHANDAKA

Tech Lead | Principal Engineer | AI Systems Architect | LangGraph · MCP · RAG

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SUMMARY

Hands-on technical leader with 21 years building production software systems. Currently architecting and implementing AI systems end-to-end — from data ingestion and embeddings to multi-agent orchestration and user-facing applications. Built two production AI systems in parallel: an AI-Powered Test Automation Framework at Entain (multi-agent orchestration, MCP servers, AI steering files) and a personal AI Stock Analyzer (14 features, 12,400+ LOC, RAG + LangGraph + Vector DB + MCP). I write code daily, architect the systems I build, and stay in the technical trenches alongside the teams I support. Fluent across the modern AI stack: Google Gemini, OpenAI, Anthropic Claude, Upstash Vector, Redis, LangGraph.

TECHNICAL DEPTH

AI Engineering (Hands-On): LangGraph (StateGraph, Conditional Routing, Multi-Agent) · RAG Pipeline Design · ReAct Agents (Tool-Calling) · MCP Server Development (Python, Node.js) · Vector Embeddings (768-dim) · Semantic Search · Prompt Engineering · LLM Tool Calling · SSE Streaming · AI Observability

LLM Models & APIs: Google Gemini (2.5 Flash, 2.0 Flash, Flash-Lite) · OpenAI GPT-4 · Anthropic Claude 3.5 · Gemini Embedding API · Multi-Model Fallback Strategies

Production AI Infrastructure: Upstash Redis · Upstash Vector DB · Multi-Key API Rotation · Rate Limit Resilience · Hybrid Caching (in-memory + persistent) · LLM Cost Optimization · Graceful Degradation

Full-Stack Engineering: TypeScript · Node.js · Express · React · Python · REST APIs · Server-Sent Events · Microservices · Vite · Vercel · Render

Test Automation Architecture: Java · Selenium WebDriver · Playwright · Cypress · Cucumber/BDD · Karate DSL · TestNG · Pytest · JMeter · Postman · TestRail · Xray · Jenkins · CI/CD

FEATURED TECHNICAL PROJECT

AI Stock Portfolio Analyzer (Personal — Production)

Live: stock-portfolio-analyzer-ten.vercel.app | **GitHub:** github.com/kmnaidu/stock-portfolio-dashboard

Production AI system I designed, built, and operate solo — 50+ days uptime, <50ms cached responses, 99%+ reliability, ₹0 infrastructure cost. Demonstrates end-to-end AI engineering depth: data ingestion → embeddings → multi-agent orchestration → user-facing responses.

Architecture Highlights:

Component	Implementation
Multi-Agent System	5 specialist agents (Analyst, Technical, Risk, News, Synthesis) with sequential orchestration + 2s rate limit delays
LangGraph Decision Agent	Conditional StateGraph (7 nodes) — short-circuits to AVOID on structural problems, saves 2 LLM calls per decision
ReAct Chat Agent	LLM tool-calling with 4 tools, 5-round reasoning loop, dynamic tool selection
RAG Pipeline	Yahoo Finance + Google RSS + Redis cache → Gemini, zero hallucination on prices
Vector DB	Gemini Embeddings (768-dim) + Upstash Vector, semantic stock similarity search
MCP Server (5 tools)	Stock data exposed via Model Context Protocol, reusable across AI IDEs
Hybrid Cache	In-memory (30s TTL) + Upstash Redis (7-day TTL), 95% hit rate
Multi-Key Resilience	4 keys × 3 models = 12 fallback attempts, 95% success on free tier
AI Observability	Every LLM call logged — model, key, timing, success, tokens
Autonomous Briefing	macOS launchd → Gemini → Twilio WhatsApp at 9:10 AM daily
SSE Streaming	Real-time word-by-word AI responses to browser

Tech Stack: React · TypeScript · Node.js · Express · LangGraph · Google Gemini · Upstash Redis · Upstash Vector · Twilio · Vercel · Render

Production Metrics: 12,400+ LOC · 76 source files · 32 stocks tracked · 50ms cached response · 95% LLM success rate · 14 features shipped

My Role: Sole architect and developer — designed system, wrote all code, deployed to production, operate it daily.

SYSTEMS DESIGNED & BUILT

System	What I Built	Technical Stack
AI Test Automation Framework (Entain)	Multi-agent system with 6 specialist agents (Ticket Analyst, Test Designer, Frontend/API Experts, Code Reviewer, Quality Gate) and MCP integration across Jira/GitLab/TestRail/Xray	Python, Node.js, MCP, Gemini, OpenAI
Multi-Agent Stock Analysis Engine	5 specialist agents with synthesis layer, sequential orchestration, rate limit resilience	LangGraph, Gemini, Node.js

System	What I Built	Technical Stack
LangGraph Decision Agent	7-node StateGraph with conditional routing for investment decisions	LangGraph, TypeScript, Gemini
RAG Pipeline	Live data retrieval + Redis cache + LLM context injection with hallucination guards	Node.js, Upstash Redis, Yahoo Finance API
MCP Server (5 tools)	Standalone server exposing stock market data to any AI IDE	Node.js, MCP SDK, TradingView API
Hybrid Cache Layer	Two-tier cache (in-memory + Redis) with automatic TTL routing	TypeScript, Upstash Redis
AI Observability	Per-call logging — model, key, timing, success, tokens, errors	Node.js, custom telemetry
Vector Embedding Pipeline	31 stock profiles → 768-dim vectors → cosine similarity search	Python, Gemini Embedding API, Upstash Vector
Multi-Key LLM Rotation	4 API keys × 3 models = 12 fallback attempts per call	Node.js, custom retry logic
Autonomous Agent (WhatsApp Briefing)	launchd-triggered → Gemini → Twilio WhatsApp delivery	Bash, Node.js, Twilio API

EXPERIENCE

Tech Lead

Entain India (formerly Ivy Comptech) — Ladbrokes, Coral Sportsbook Sep 2019 – Present | Hyderabad | 6 yrs 10 mos

Hands-On Technical Contributions:

- Architected an AI-Powered Test Automation Framework integrating multi-agent orchestration, MCP servers, AI steering files, and event-driven hooks
- **Wrote the core MCP servers** for Jira Cloud, GitLab API, TestRail, Xray, and a custom Project Bridge — enabling AI to search code across repos, create branches, commit code, and manage workflows
- **Designed and implemented** specialist AI agents in Python and Node.js: Ticket Analyst, Test Designer, Frontend Expert, API Expert, Code Reviewer, Quality Gate
- **Built the AI Steering Files system** — persistent context layer encoding team conventions, loaded conditionally per file type

- **Configured AI Hooks** — event-driven automation for quality gates, linting, Gherkin validation, and post-task test execution
- Architected end-to-end test frameworks in Java, Playwright, Selenium, Cypress, Cucumber BDD, and Karate DSL
- Established CI/CD pipelines with Jenkins, BrowserStack, LambdaTest integration, and TestRail reporting

Technical Impact:

- Test case design time: 2-3 hours → 15-30 min (80% faster, ~40 hours/week saved)
- Code generation: 4-6 hours → 30-60 min (75% faster, 2x feature delivery)
- Cross-repo search: 30+ min → <30 sec (98% faster)
- Regression coverage: 80% automated, enabled monthly → weekly releases

Team Context: Tech lead role within a 10-engineer team (managed teams up to 20 prior to recent restructuring). Hands-on coding alongside mentoring, while owning architecture decisions for the AI framework.

Lead Engineer

Scientific Games Pvt Ltd (WMS) | Aug 2012 – Aug 2019 | Pune | 7 yrs 1 mo

RGS Platform — Casino Games platform with regulatory compliance requirements

- Hands-on automation framework design for gaming platform products
- **Built automation scripts** using Java, TestNG, JMeter for regression and performance testing
- **Automated COAT back office** end-to-end with Selenium WebDriver + TestNG
- **Developed performance testing infrastructure** using JMeter with custom Beanshell scripts
- Authored core utility libraries reused across multiple gaming product lines

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Test Lead

AgreeYa Mobility | Nov 2011 – Aug 2012 | Pune | 10 mos

- Built BDD test automation using Cucumber + Selenium Ruby for Vendor Self Invoicing System
- Authored reusable step definitions and page objects
- Designed test data fixtures and scenario outlines

Senior Software Engineer (Automation Test Lead)

Motorola Mobility | Apr 2010 – Nov 2011 | Hyderabad | 1 yr 8 mos

- **Built Invader+ automation framework** from scratch using Selenium WebDriver
- Automated 40%+ of browser compliance test suite
- Integrated new feature test coverage into existing framework

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Earlier Technical Roles (2005 – 2010)

- **Azingo Inc.** (Senior QA Engineer) — JavaScript test development for mobile web widgets across multiple platforms
- **Mascon Global / Motorola Client** (Senior Software Engineer) — Bluetooth connectivity testing, shell script development for voice/data validation
- **Tata Elxsi Ltd** (Test Engineer) — Bluetooth interoperability and stress testing across device portfolio

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EDUCATION

MCA (Master of Computer Applications) — Andhra University | 2001 – 2004 **B.Sc Computer Science** — Andhra University | 1998 – 2001

CERTIFICATIONS & LEARNING

- Generative AI: Working with Large Language Models (LinkedIn Learning, 2025)
- LangGraph & Multi-Agent Systems (production implementation)
- Model Context Protocol (MCP) — server development and enterprise integration
- RAG Architecture & Vector Embeddings (production implementation)

LANGUAGES

English (Fluent — Business/Technical) · Telugu (Native) · Hindi (Conversational)

OPEN-SOURCE & PUBLICATIONS

- **GitHub:** github.com/kmnaidu — Stock Portfolio Analyzer (full production code)
- **Published:** MCP server for Indian stock market data (reusable across AI IDEs)
- **LinkedIn Articles:** 5 technical posts on AI Engineering patterns (3,800+ impressions, 80+ reactions)
- **Medium:** "How I Built an AI Stock Portfolio Analyzer Using RAG, Multi-Agents, LangGraph, Vector DB, and Redis"